

LITERATURE SURVEY

Aim: LITERATURE SURVEY ON ARTIFICIAL INTELLIGENCE

Paper 1:

Title: Intelligent Optimization Under Multiple Factories: Hybrid Flow Shop Scheduling Problem with Blocking Constraints Using an Advanced Iterated Greedy Algorithm.

Authors: YONG WANG, YUTING WANG, YUYAN HAN, JUNQING LI, KAIZHOU GAO , YUSUKE NOJIMA.

Summary:

The distributed hybrid flow shop scheduling problem (DHFSP) combines distributed production environments with parallel machine settings. However, in practical manufacturing systems, buffers are often unavailable between certain consecutive machines, which leads to blocking situations. To better capture real-world production scenarios, this study investigates the distributed hybrid flow shop scheduling problem with blocking constraints (DBHFSP). A mixed integer linear programming (MILP) formulation is developed for the problem, and its correctness is verified using the Gurobi optimization solver.

To efficiently solve the DBHFSP, the paper proposes an advanced iterated greedy (AIG) algorithm aimed at minimizing the makespan. The classical Nawaz–Enscore–Ham (NEH) heuristic is adapted to accommodate blocking conditions. Additionally, two inter-factory neighborhood search mechanisms and a swap-based local search strategy are incorporated to enhance both global exploration and local exploitation. Considering that factories operate independently, a memory-based decoding approach is introduced to accelerate insertion operations and reduce computational effort.

To avoid premature convergence, two distinct shaking strategies are embedded in the algorithm. The performance of the AIG algorithm is evaluated against five state-of-the-art methods on 80 benchmark instances. Computational experiments demonstrate that the proposed AIG algorithm consistently outperforms the competing approaches, achieving a 1.0% reduction in makespan and an 86.1% improvement in relative percentage increase (RPI), thereby confirming its effectiveness for solving the DBHFSP.

Paper 2:

Title: Sensor Selection With Cost Function Using Nondominated-Solution-Based Multiobjective Greedy Method

Authors: YUJI SAITO, KUMI NAKAI, TAKAYUKI NAGATA, KEIGO YAMADA, TAKU NONOMURA, KAZUKI SAKAKI, YASHIO NUNOME.

Summary:

This work presents a novel cost-constrained greedy algorithm for sensor selection derived from a nondominated-solution-based multiobjective greedy (NMG) framework. The proposed approach is evaluated against an existing sensor selection technique. By jointly considering the sensing cost and the D-optimality criterion, the algorithm identifies sensor subsets through the concept of nondominated solutions rather than optimizing a single aggregated objective.

Earlier multiobjective sensor selection methods with cost limitations typically relied on weighted linear combinations of objectives, which required tuning a hyperparameter to control the trade-off between performance and cost. In contrast, the proposed NMG approach eliminates the need for such parameter tuning and is capable of directly generating Pareto-optimal solutions.

The performance of the proposed algorithm is validated using three real-world datasets, including sea surface temperature measurements, airfoil flowfield data, and rocket combustion chamber simulations. A binary cost is assigned to each candidate sensor location, and sensor selection under a fixed cost budget is examined. Experimental results indicate that the NMG algorithm reliably yields Pareto-optimal sensor configurations, with the majority of selected sensor sets achieving better objective values than those obtained by the benchmark method at equivalent cost levels.

Paper 3:

Title: Efficient Densest Subgraphs Discovery in Large Dynamic Graphs by Greedy Approximation

Authors: TAO HAN.

Summary:

Identifying densest subgraphs is a fundamental problem in graph mining and plays an important role in tasks such as community detection and event identification within large-scale, evolving networks. Although several techniques have been developed for large or dynamic graphs, effective solutions that simultaneously address both scale and continuous changes are still limited. This study formulates the densest subgraph detection problem as NP-hard and introduces an incremental greedy approximation algorithm with a time complexity of $O(m+n)O(m+n)$.

The proposed approach tackles the problem by heuristically combining locally dense subgraphs. The algorithm operates in four main phases. First, the edge stream of a dynamic graph is partitioned into multiple subgraphs over a specified time window TTT . Next, locally dense candidate subgraphs are identified to form a local candidate set. These candidates are then heuristically merged to generate global densest subgraph candidates. Finally, a constrained static densest subgraph detection algorithm is applied to these candidates to extract the final densest subgraphs.

Compared with existing methods that are primarily designed for small and static graphs, the proposed technique significantly enhances scalability. Experimental evaluations on real-world networks containing billions of nodes demonstrate that the method reduces execution time by approximately 25% relative to static approaches, while also producing more accurate and compact representations of graph structures. Moreover, the algorithm performs robustly across graphs with diverse density distributions, highlighting its suitability for highly dynamic environments.

Analysis Table:

Title of paper	Advantages	Disadvantages	Open Challenges
Intelligent Optimization Under Multiple Factories: Hybrid Flow Shop Scheduling Problem with Blocking Constraints Using an Advanced Iterated Greedy Algorithm	The AIG algorithm effectively minimizes makespan while incorporating blocking constraints, outperforms five advanced algorithms, and demonstrates significant improvements in computational efficiency and solution quality.	The approach heavily relies on computational resources, and its performance may vary with problem size and complexity, potentially limiting scalability in larger systems.	Addressing dynamic scheduling environments, incorporating real-time adjustments, and improving scalability for large-scale distributed systems remain key challenges.
Sensor Selection With Cost Function Using Nondominated-Solution-Based Multiobjective Greedy Method	The NMG algorithm effectively identifies Pareto-optimal sensor sets without requiring hyperparameters, consistently outperforming previous methods in terms of cost-efficiency and objective values.	Its performance may be dataset-dependent, and the binary cost function could limit flexibility in modeling more complex cost constraints.	Extending the approach to handle dynamic, multi-sensor environments and incorporating non-binary cost functions or additional real-world constraints remain key areas for future research.
Efficient Densest Subgraphs Discovery in Large Dynamic Graphs by Greedy Approximation	The proposed incremental greedy method efficiently handles large, dynamic graphs with reduced running time (25% faster) and provides accurate estimations of compact subgraphs, improving scalability over static methods.	The heuristic approach may not guarantee optimal solutions and could face limitations in extremely high-density or rapidly changing graphs.	Extending the method to support real-time subgraph updates and addressing the trade-off between approximation accuracy and computational efficiency in highly dynamic graph environments.

Conclusion:

The studies collectively highlight advancements in solving complex optimization problems across different domains. For distributed hybrid flow shop scheduling, the Advanced Iterated Greedy (AIG) algorithm effectively minimizes makespan under blocking constraints, demonstrating significant improvements over other methods. In sensor selection, the Nondominated Multiobjective Greedy (NMG) algorithm offers hyperparameter-free Pareto-optimal solutions, outperforming prior approaches in cost-constrained scenarios. For densest subgraph detection, the proposed incremental greedy method excels in scalability and accuracy for large, dynamic graphs. These methods emphasize the importance of computational efficiency, scalability, and adaptability in addressing real-world challenges, paving the way for future research to tackle dynamic, large-scale, and constraint-laden problems.



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